

# ELECTRONIC TOLL COLLECTION (ETC) ADMIN PORTAL

## SYSTEM OPERATIONAL USER MANUAL

**Document Version:** 1.0.0 (Production)

**Last Updated:** September 2026

**Target Audience:** System Administrators, Operation Managers & Branch Officers

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## 1. OVERVIEW OF THE SYSTEM

The **ETC Admin Portal** is an enterprise-grade back-office management application designed to clear high-volume electronic toll transactions (Toll Operations) and recharge entries (TopUp System). The system coordinates directly with the central API Gateway (**ETCGatewayAPI**) and safely commits batch records to the Core Banking System (CBS).

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## 2. ARCHITECTURE COMPONENT MAP

- **ETCGatewayAPI:** The primary microservice gateway routing all payload actions.
  - **EtcMwApi:** Core Business Logic engine managing data validations.
  - **ETCAdminUI:** The web dashboard interface used by operators to perform administrative actions.
  - **Etc.Shared:** Dynamic library syncing shared Contracts and DTO instances.
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## 3. CORE MODULE INSTRUCTIONS

### A. Login & Identity Session

1. Access the web endpoint URL via an active secure browser session.
2. Enter your assigned **User ID**, secure **Password**, and select your routing branch setup.
3. System sessions auto-expire after **30 minutes of idle time** to protect administrative compliance integrity.

### B. Data Processing & Clearing Panel

To run clearing settlement procedures, navigate to the **Settlement Process** link from the header block:

1. **Select Parameters:** Choose the operational target date using the date selector and choose between **Toll Processing** or **TopUp Processing**.
2. **Execute Stage 1 (Run Data Process):** Click the button to group pending transaction states. The system will auto-generate a unique **BatchProcessId** and calculate gross, reversed, and net clearable figures.
3. **Execute Stage 2 (Post Settlement):** Click the action button to fire the structured accounting payload payload directly to the CBS framework.
  - *Toll Mechanism:* Executes a Double-entry voucher debiting the **ETC Parking GL** and crediting the target **Settlement Account**.
  - *TopUp Mechanism:* Transitions transaction record states directly into a cleared profile without secondary GL postings.

### C. System Security (Password Rotation)

1. Go to the User Context menu drop-down or click **Change Password**.
  2. Provide your active **Current Password** in the encrypted input block.
  3. Type the **New Password** confirming to password complexity standards and save.
  4. *Recommendation:* Rotate your core administrative keys every **90 days**.
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## 4. TELEMETRY STATUS CODES

When running settlement processes, the AJAX alert module returns standard infrastructure status codes:

- **200 OK** – Settlement pipeline executed flawlessly.
  - **404 NOT FOUND** – No active transaction records matched the target filter parameters for that date.
  - **409 CONFLICT** – Data has already been processed or is currently being modified by another operator session.
  - **500 SERVER ERROR** – Database context connectivity dropped or internal proxy failure occurred. Review system error logs immediately.
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## 5. HELP & ESCALATION SUPPORT

For internal application blockages or critical API exceptions:

- **Technical Team:** Internal Enterprise Network Infrastructure Division
- **Log Registry:** Review Serilog output pipes defined in the `appsettings.json` diagnostics target.